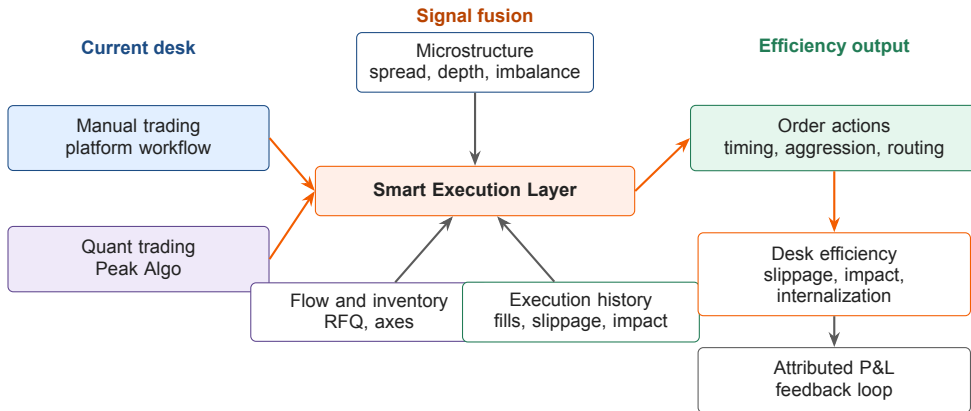


Liquidity-Aware Smart Execution for Desk Efficiency

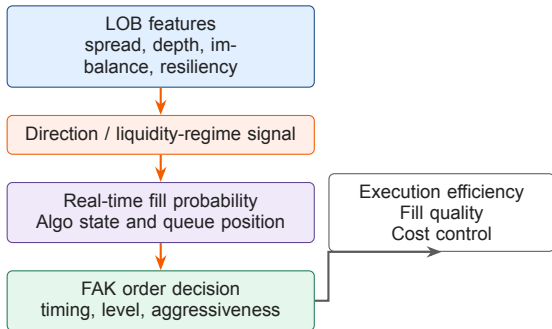
Microstructure Signals Across Manual Trading and Peak Algo

Yida Ding | June 2026

■ Smart Execution as a Desk Efficiency Layer



■ FAK Order Execution Efficiency



Case 1: favorable direction

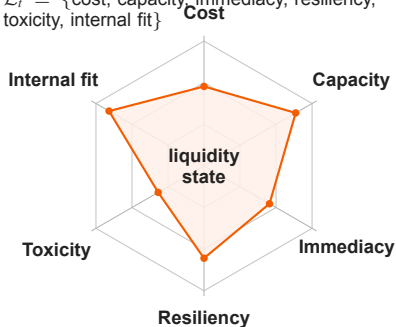
- ▶ Direction supports the order side.
- ▶ Expected fill probability is higher.
- ▶ Execution efficiency: lower slippage and lower market impact.

Case 2: adverse direction

- ▶ Direction works against the FAK order.
- ▶ Revise fill expectation and aggressiveness.
- ▶ Improve success rate and reduce unhedged risk.

■ Liquidity is the True State Variable

$\mathcal{L}_t = \{\text{cost, capacity, immediacy, resiliency, toxicity, internal fit}\}$



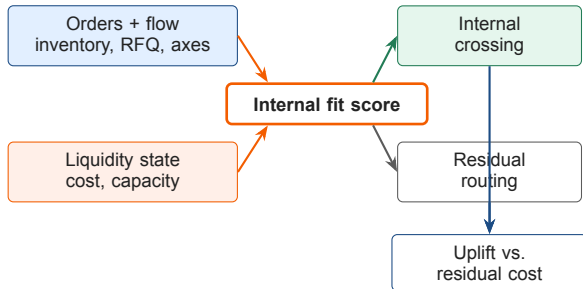
How to read the state

- ▶ **High capacity + fast resiliency:** raise FAK confidence.
- ▶ **High cost or toxicity:** revise fill expectation and aggression.
- ▶ **High internal fit:** cross internally before external routing.
- ▶ **Weak state:** delay, slice smaller, or hedge earlier.

Design rule

Price is the observation. Liquidity is the control state.

■ Microstructure for Internal Flow



Research-backed decision

- ▶ Price external liquidity by state, not quote alone.
- ▶ Estimate internal fit from inventory, RFQ and axes.
- ▶ Route residual flow when depth and resiliency are favorable.
- ▶ Prove uplift against residual external cost.

■ 0.01 bp as Attributed Execution P&L

Annual efficiency target

90M

annual target from 0.01 bp saving

- ▶ 2025 flow: 45tn.
- ▶ Average duration: 2 years.
- ▶ $45tn \times 2 \times 0.01bp \approx 90M$.

Attribution stack

- ▶ **Internal crossing:** avoid spread and external market impact.
- ▶ **Liquidity timing:** trade when depth and resiliency are favorable.
- ▶ **Adverse-selection control:** reduce hedge leakage and bad fills.

Proof metrics

- ▶ Slippage vs. arrival / decision price.
- ▶ Post-trade impact and hedge delay.
- ▶ Internalization uplift vs. residual external cost.